

Low noise amplifier with active feedback structure for implantable neural recording

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Abstract: A novel low-power, low-noise amplifier is described for biomedical and bioengineering brain neural recording applications. The bioamplifier achieves the best power-size tradeoff compared to the previous design. By means of a new active feedback configuration, the DC offset is rejected without the large capacitors. An active differentiator with an amplifier in the feedback path places a high-pass cutting frequency in the transfer function. The midband gain consists of the passive components, and is insensitive to the mismatch of process. The bioamplifier has been implemented in the 0.35- μm 2P4M CMOS process and occupies 0.052 mm² of die area. The current consumption of amplifier is 2 μA at $\pm 1.5\text{ V}$ supply. The bioamplifier achieves a midband gain of 42 dB and a -3 dB bandwidth from 13 Hz to 10.5 kHz. The input-referred noise is 5.7 μV_{rms} corresponding to an NEF of 3.1.

Keywords: bioamplifier, active feedback, low noise, DC-offset

Classification: Integrated circuits

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1 Introduction

The clinicians demand a system to record the simultaneous activities of large numbers of neurons in the brain for understanding how the brain works. Neural recording from the large-scale chronic multi-electrodes has been widely used in many biomedical applications [1, 2, 7, 9]. The bioamplifier is one of the most important parts in this system.

Neural signals from extracellular recording are very weak. The amplitude is in 50–500 μV range with the noise level in 5–10 μV . Such signals cannot be processed further without amplification. To increase the SNR of the system, the input referred noise of the amplifier need to minimize. However, the designers meet the challenge about the tradeoff between low-noise and low-power designs. The multi-electrodes recording system incorporates a large number of neural amplifiers. Too much power dissipated by the amplifiers may damage the surrounding issues by heating. The electrode shape restricts the size of the each amplifier as low as 0.16 mm² [1]. Due to the electrode potential mismatch, the DC offset up to 1 V is cross the recording electrodes. The offset which is orders of magnitude larger than the biopotential signal, must be rejected to prevent the saturation of the amplifier. The power, noise, size and DC offset are the major problems faced by designers.

A large amount of neural amplifiers have been presented in the literature. There are few DC offset suppression schemes adopted for on-chip integration in neural recording system. First method that uses a capacitor feedback network with ac coupling of input devices to reject the offset is very popular in designs [1, 3, 5]. The MOS-bipolar pseudo-resistor component in parallel with a capacitor feedback provides the very small low-cutoff frequency. The midband gain is the ratio of feedback capacitors. The major drawback is this configuration requires very large capacitors which occupy great percents area (> 60%) of chip. The second method employs a closed-loop resistive feedback and electrode capacitance to form a highpass filter [4, 6]. The performances of this approach depend on the electrode parameters which change from one site to another. The third method adopts the active low-frequency suppression that an active integrator in the feedback path of low noise amplifier forms a highpass cutoff frequency response [2]. This technology achieves small size without large passive devices. However, the amplifier works with the open-loop mode in the midband, and the gain suffered from the variation of the open-loop gain of low noise amplifier [8].

This paper presents a bioamplifier by means of a novel active DC offset suppression which takes the merits of capacitors feedback network configuration [1] and active low-frequency suppression configuration [2]. The midband gain is determined by the ratio of the capacitors, and size of capacitors is similar to the active cutoff-frequency suppression configuration without increasing the power and noise.

2 Architecture of bioamplifier

The neural amplifier must be as optimizing as possible to satisfy the performance requirement of low noise and the tolerance of DC offsets voltage. In order to implement the trade-off between the amp gain and noise, a medium gain and lowest noise in the first stage is necessary to keep the noise contributions from the following stages negligible. In addition, the DC offsets rejection is suppressed by the first stage architecture.

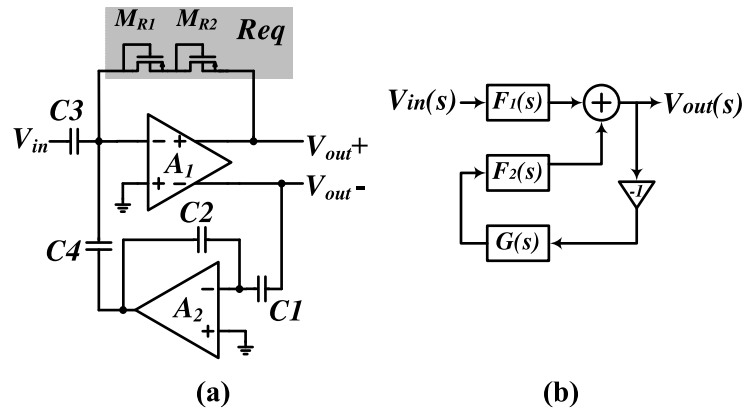


Fig. 1. (a) Schematic of the bioamplifier and (b) equivalent feedback control

2.1 Amplifier with active feedback

Fig. 1 (a) shows a schematic of our bioamplifier design with the proposed with DC offset suppression configuration. This design consists of a differentiator and a close-loop amplifier within its feedback path. The differentiator is made from an amplifier (A_1), capacitors (C_3 , C_4), and a high-value resistor (R_{eq}). A close-loop amplifier is composed of an amplifier (A_2), and feedback capacitors (C_1 , C_2). The differentiator is equivalent to a highpass filter to reject the DC offset. The close-loop amplifier in feedback path corrects the highpass function of the differentiator to form the stable midband gain.

Fig. 1 (b) shows a functional block diagram of the bioamplifier which consists of differentiator F_1 , F_2 and amplifier G with capacitors feedback network. In the case of all the amplifiers are ideal with no parasitic capacitance, the transfer function of the ideal bioamplifier is

$$\frac{V_{out}(s)}{V_{in}(s)} = \frac{F_1}{1 - F_2 \cdot G} = -\frac{sR_{eq}C_2C_3}{sR_{eq}C_1C_4 + C_2} \quad (1)$$

where $F_1(s) = -sR_{eq}C_3$, $F_2(s) = -sR_{eq}C_4$, $G(s) = -C_1/C_2$.

The midband gain $A_M = -C_2C_3/C_1C_4$ is comprised of two capacitive ratios. The gain of the new bioamplifier architecture equals two previous amplifiers working in series. For a given gain, the bioamplifier demands smaller capacitors. The transfer function reveals the -3 dB cutoff frequency to be

$$f_H = \frac{1}{2\pi} \frac{C_2}{R_{eq}C_1C_4} = \frac{1}{2\pi} \frac{A_M}{R_{eq}C_3} \quad (2)$$

To achieve -3 dB highpass cutoff frequency around 10 Hz would requires the resistor R_{eq} in the order of $10^{12} \Omega$. Transistors M_{R1} – M_{R2} are MOS-bipolar devices which act as pseudo resistors [1]. For small input amplitude, the equivalent resistor is more than $10^{13} \Omega$.

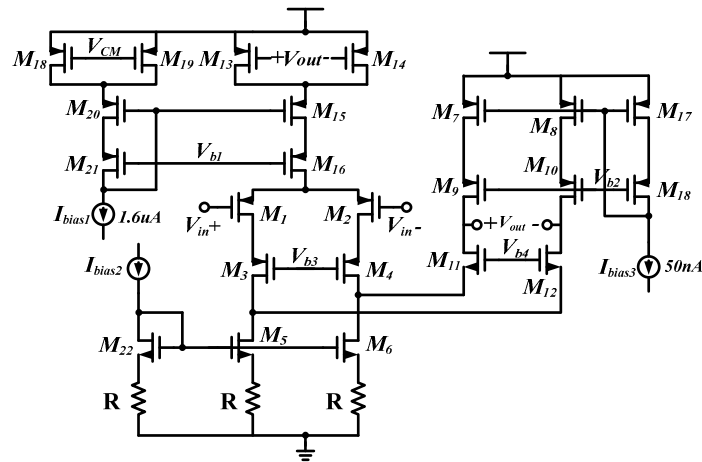


Fig. 2. Circuit schematic of the low power, low noise amplifier A₁

The amplifier A_1 has a folded-cascode topology with source-degenerated current mirrors [3] showed in Fig. 2. The full differential configuration which has the high CMRR to reject interference is adopted. The common mode feedback circuit is formed by the transistor M_{13} - M_{14} in triode region to sense the CM level. The DC gain of A_1 should satisfy the relationship given in (6). For $\alpha = 0.95$, the gain cannot be low than 72 dB. The -3 dB bandwidth of A_1 is determined by neural signal bandwidth. Simulations show that the amplifier A_1 has a DC gain of 124 dB, and a unit-gain bandwidth of 3.6 MHz.

2.2 Effect of amplifier finite gain

The finite gain of the amplifier (A_1 and A_2) donated by A_{v1} and A_{v2} , respectively. C_{in1} and C_{in2} are the input parasitic capacitors of the amplifier. For the case where $A_{v2}C_2 \gg C_1 + C_2 + C_{in2}$ and $A_{v1} \gg 1$, the transfer function is

$$\frac{V_{out}(s)}{V_{in}(s)} \approx -\frac{sR_{eq}A_{v1}C_2C_3}{sR[A_{v1}C_1C_4 + (C_3 + C_4 + C_{in1})C_2] + A_{v1}C_2} \quad (3)$$

When A_{n1} satisfies the following relationship

$$A_{v1} \gg \frac{(C_3 + C_4 + C_{in})C_2}{C_1 C_4} \quad (4)$$

the midband gain of bioamplifier is the same as ideal condition. If A_{v1} fails to satisfy (4), $|V_{out}(s)/V_{in}(s)|$ will be smaller than C_2C_3/C_1C_4 . In order to characterize the effect of finite gain on the midband gain, a parameter α is defined as

$$\left| \frac{V_{out}(s)}{V_{in}(s)} \right| = \alpha |A_M| \quad (5)$$

The actual midband gain falling into α times of ideal value requires the gain A_{v1} to be

$$A_{v1} \geq \frac{1}{1/\alpha - 1} |A_M| \quad (6)$$

Considering the A_1 as a simple first-order model for the transfer function of a dominant-pole compensated amplifier, $A_{v1}(s)$, is given by

$$A_{v1}(s) = \frac{A_{v1,open}}{1 + s/\omega_0} \quad (7)$$

Where $A_{v1,open} = G_m R_{out}$ is the DC gain of amplifier A_1 , and $\omega_0 = 1/(R_{out} C_L)$ is the dominant pole. The -3 dB lowpass cutoff frequency can be evaluated from (3)

$$f_L = \frac{1}{2\pi} \frac{g_m}{C_L} \frac{C_1 C_4}{C_2 C_3} = \frac{1}{2\pi} \frac{g_m}{C_L} |A_M| \quad (8)$$

The -3 dB lowpass cutoff frequency depends on the transconductance g_m of A_1 , load capacitor C_L and midband gain A_M of bioamplifier. The gain A_{v2} of amplifier A_2 has no effect on the midband gain in the case of $A_{v2} \gg (C_1 + C_2)/C_2$ which can achieve easily.

2.3 Effect of noise

In the previous section, the transfer function is deduced under ideal conditions. The amplifier finite DC gain, parasitic capacitances and noise are devoted to errors that alter the transfer function of bioamplifier. Equivalent model of these non-ideal factors in the circuits is shown in Fig. 3.

Noise sources associated with the amplifier A_1 and A_2 are represented as V_{n1} and V_{n2} , respectively. Thermal noise source of the equivalent resistor R_{eq} is V_R , as shown in Fig. 3. Consider the equivalent noise model of Fig. 3, the overall transfer function from noise source to amplifier output can be written as

$$\frac{V_{out}}{V_{n1}} = \frac{A_1(sR_{eq}M + 1)(A_2C_2 + N)}{sR_{eq}A_1A_2C_1C_4 + (sR_{eq}M + 1)(A_2C_2 + N) + A_1(A_2C_2 + N)} \quad (9)$$

$$\frac{V_{out}}{V_{n2}} = -\frac{sR_{eq}A_1A_2NC_4}{sR_{eq}A_1A_2C_1C_4 + (sR_{eq}M + 1)(A_2C_2 + N) + A_1(A_2C_2 + N)} \quad (10)$$

$$\frac{V_{out}}{V_R} = -\frac{A_1(A_2C_2 + N)}{sR_{eq}A_1A_2C_1C_4 + (sR_{eq}M + 1)(A_2C_2 + N) + A_1(A_2C_2 + N)} \quad (11)$$

If $A_2C_2 \gg N$, $A_1 \gg 1$, $A_1C_1C_4 \gg MC_2$, and $s \gg 1$, then, equivalent input transfer function from noise source is simply

$$\begin{aligned} \frac{V_{in}}{V_{n1}} &= \frac{V_{out}}{V_{n1}} \cdot \frac{V_{in}}{V_{out}} = \frac{(C_3 + C_4 + C_{in1})C_2}{C_1C_4} \cdot \frac{C_1C_4}{C_2C_3} \\ &= \frac{C_3 + C_4 + C_{in1}}{C_3} \end{aligned} \quad (12)$$

$$\frac{V_{in}}{V_{n2}} = \frac{V_{out}}{V_{n2}} \cdot \frac{V_{in}}{V_{out}} = -\frac{(C_1 + C_2 + C_{in2})C_2}{C_1} \cdot \frac{C_1C_4}{C_2C_3}$$

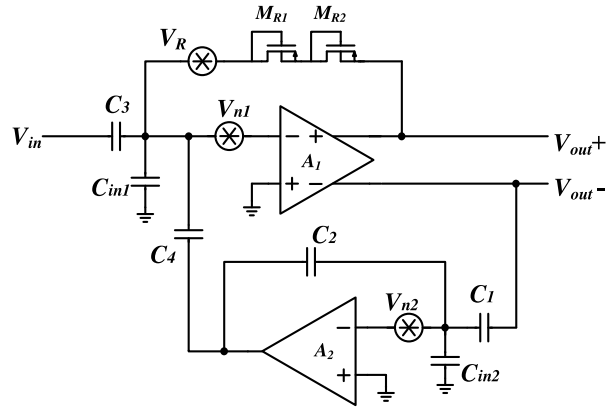


Fig. 3. Equivalent noise model for evaluating the effect of non-ideal factors to bioamplifier.

$$= -\frac{(C_1 + C_2 + C_{in2})C_4}{C_2C_3} \quad (13)$$

$$\frac{V_{in}}{V_R} = \frac{V_{out}}{V_R} \cdot \frac{V_{in}}{V_{out}} = \frac{C_2}{sR_{eq}C_1C_4} \cdot \frac{C_1C_4}{C_2C_3} = \frac{1}{sR_{eq}C_3} \quad (14)$$

The input referred noise power of the bioamplifier can be obtained, as follows

$$\begin{aligned} \overline{V_n^2} \approx & \left(\frac{1}{sR_{eq}C_3} \right)^2 \cdot \overline{V_{n,R}^2} + \left(\frac{C_3 + C_4 + C_{in1}}{C_3} \right)^2 \cdot \overline{V_{n,1}^2} \\ & + \left[\frac{(C_1 + C_2 + C_{in2})C_4}{C_2C_3} \right]^2 \cdot \overline{V_{n,2}^2} \end{aligned} \quad (15)$$

The flick noise limits the performance of the low power, low frequency amplifier. In order to minimize the flick noise, the input devices of amplifier are PMOS transistors with large area. The large input transistors increase the input capacitance, and decrease the noise performance as show in (15). The design of amplifier A₁ needs to take care of the tradeoff between the flick noise and input capacitance. The noise of amplifier A₂ has less contribution to the input referred noise of bioamplifier. For the case where C₂ = C₃ = 10C₁ = 10C₄, C_{in1} = C_{in2}, the input referred noise related noise source V_{n2} is

$$\overline{V_{n,in2}^2} = \frac{1}{100} \left(\frac{C_3 + C_4 + C_{in1}}{C_3} \right)^2 \cdot \overline{V_{n,1}^2} = \frac{1}{100} \overline{V_{n,in1}^2} \quad (16)$$

The gain and noise of A₂ have less effect on bioamplifier that allows us to relax the requirement of design for saving power. Integrated the input referred noise given by R_{eq} is

$$V_{n,inR}^2 = \int_{f_L}^{f_H} \left(\frac{1}{2\pi f R_{eq} C_3} \right)^2 \overline{V_{n,R}^2} df = \frac{\overline{V_{n,R}^2}}{2\pi R_{eq} C_3} \frac{f_H - f_L}{f_H f_L} \quad (17)$$

The noise of equivalent resistor R_{eq} undergoes attenuation in the bioamplifier. Therefore, the input referred noise of bioamplifier is donated by A₁ only.

2.4 Analysis of stability

As depicted in Fig. 4(a), the magnitude of differentiator increases with 20 dB/decade and its phase is −90°. The non-ideal differentiator presents

a finite gain at the high frequency that equal to the gain A_{v1} of amplifier A_1 . The differentiator low corner frequency is at $\omega = A_{v1}/(R_{eq} \cdot C)$. Notice that the A_{v1} must be as large as possible to fade non-ideal effects in differentiator. Assume the τ_p is the dominant pole of the amplifier A_1 . The differentiator has the high corner frequency of τ_p . As the frequency is high than τ_p , the gain of differentiator decreases and the phase is -180° . By this reason, the τ_p should be large than the neural signal bandwidth.

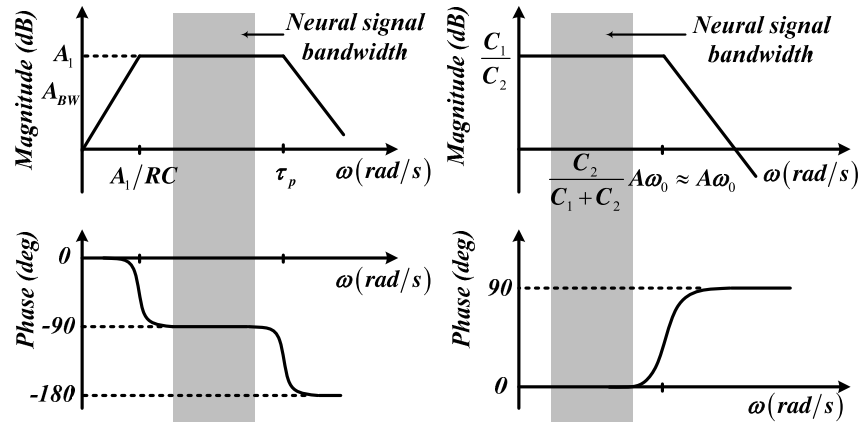


Fig. 4. Magnitude and phase responses of (a) the differentiator and (b) the amplifier with capacitors feed-back network.

The block $G(s)$ which consisted of the amplifier A_2 and capacitors feed-back network as show in Fig. 1 (b) has the DC gain $(-C_1/C_2)$. In the case of the amplifier is the simple pole configuration, the open-loop gain is A_{v2} , open and the main pole is ω_1 . The block G has the corner frequency that is equal to the gain-bandwidth product $(A_{v2,open} \cdot \omega_1)$ and the phase is 90° . The neural signal bandwidth must be within this corner frequency.

The proposed bioamplifier configuration presents a phase shift of 180° between node $+V_{out}$ and $-V_{out}$. The differentiator F_2 (Fig. 1 (b)) generates a negative phase shift of -90° in loop gain. The block G introduces a positive phase shift φ_G . The φ_G equals 90° under the assumption that the amplifier A_2 has only the dominant pole ω_1 . If the A_2 has the other pole high than ω_1 , the phase shift φ_G will be large than 90° . Therefore, the total phase shift introduced in the loop gain is $\varphi_{LG}(j\omega) = 180^\circ - 90^\circ + \varphi_G$. At unity-gain frequency, the phase margin (PM) of the whole bioamplifier is

$$PM = \varphi_{LG}(j\omega_t) - 180^\circ = \varphi_G - 90^\circ \quad (18)$$

Hence at ω_t , a phase shift of $\varphi_G = 90^\circ$ yields a PM of zero. Where a phase shift of $\varphi_G > 90^\circ$ yields the $PM > 0$.

3 Implementation of bioamplifier

The midband gain of the bioamplifier is the product of C_2/C_1 and C_3/C_4 . If the gain is constant, decrease the total capacitance demands minimum the

C_1 and C_4 . However the parasitic capacitance restricts the minimum value of C_1 and C_4 . The total size of capacitor is minimum under the relation of $C_2/C_1 = C_3/C_4$. This scheme is not the optimization of noise. Increase C_3/C_4 can attenuate the noise with V_{n1} as given in (15). The optimal value of C_3/C_4 is select to achieve the minimum noise-capacitor product for the bioamplifier. With midband gain of 200, the normalization of noise-capacitor product varies with C_3/C_4 is show in Fig. 5.

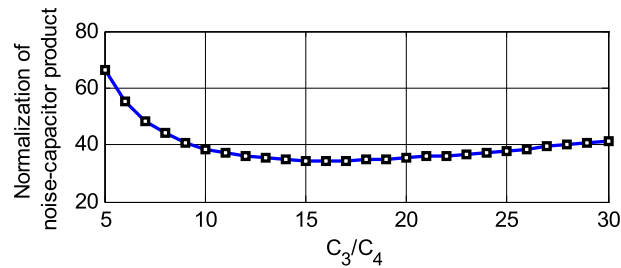


Fig. 5. The normalization of noise-capacitor product varies with C_3/C_4 .

The noise-capacitor product has the minimum value of $C_3/C_4 = 16$ by setting the value of C_3 to 1.6 pF, C_2 to 1.25 pF, C_1 and C_4 to 100 fF using the poly-poly capacitors. The total capacitance of the full differential scheme is only 6.1 pF. As stated above, the finite DC gain, noise and bandwidth of amplifier A_2 have less impact on the bioamplifier. The specifications can be relaxed for A_2 to save the power. A_2 has the same configuration as A_1 except using the common current mirrors instead of the source-degenerated current mirrors. The DC gain of A_2 is 136 dB. The bandwidth is 892 kHz and the phase margin is 57° , while consuming 300 nA.

We simulated the effect of different substrate bias voltage on the threshold voltage and noise current. Fig. 6 shows the substrate to source bias voltage (V_{BS}) dependence of $1/f$ noise for normalized drain current noise power spectral density (S_{id}) and depletion capacitance. In the Fig. 6, S_{id} has decreased by 50% if $V_{BS} = 0.5$ V. Because of depletion layer width enhancing, the

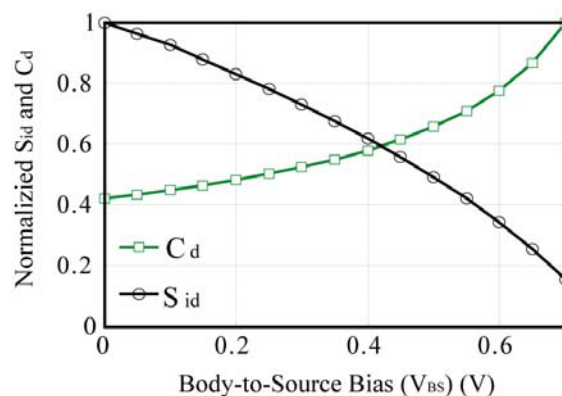


Fig. 6. Normalized S_{id} and C_d versus body-to-source bias voltage.

space between the carrier and the trap increasing, the probability of the carrier capture decrease, then, it affect $1/f$ noise.

The substrates of the NMOS and PMOS are usually connected to most negative and positive rails, respectively, to avoid latch-up. The threshold voltage decrease 100 mV (NMOS) and 130 mV (PMOS) if V_{BS} increase to 0.5 V from 0 V. The PN-junction between source and substrate become turn-on with increasing V_{BS} , then, it can produce latch-up. In this paper, the bias voltage of substrate of NMOS and PMOS is set less than 0.4 V to avoid latch-up.

4 Experiments result

The bioamplifier is designed and verified using Spectre simulator and virtuoso layout design platform. The chip are implemented using a technology library of $0.35\ \mu\text{m}$ 2-poly 4-metal CMOS process and the chip photo is shown in Fig. 7. The chip is arranged in 16-channel array including bioAmp, detector [10], and ADC block. Test results of amplifier have been performed and are summarized in Table I. The bioamplifier with full differential scheme fits in $275\ \mu\text{m} \times 190\ \mu\text{m}$ area.

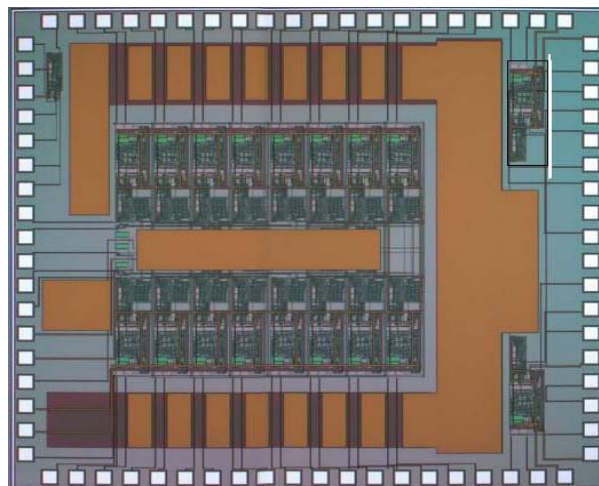


Fig. 7. Microphotograph of 3×3 mm chip including 16-channel of bioamplifier.

Figure 8 presents the measured frequency and phase responses. The mid-band gain of 42 dB and the bandwidth from 13 Hz to 10.5 kHz have been measured. The cutoff frequencies f_{hp} and f_{lp} are located at 13 Hz and 10.5 kHz respectively, which yields a bandwidth of 10 kHz. Fig. 9 shows a plot of the measured and simulated input-referred noise. The thermal noise level is about $56\ \text{nV}/\sqrt{\text{Hz}}$. Integrated noise from 13 Hz to 10.5 kHz yields an RMS noise of $5.7\ \mu\text{V}_{\text{rms}}$. Table I shows the state of the art in bioamplifier. The noise efficiency factor (NEF) is a measure of the noise performance of a given bioamplifier as a function of its power consumption and the bandwidth. Base on the simulation results, the NEF of this bioamplifier is found to be 3.1.

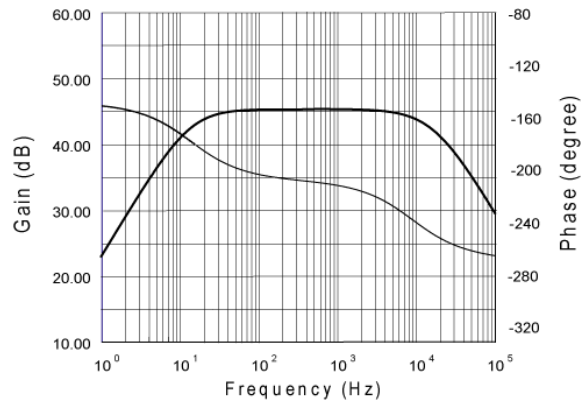


Fig. 8. Measured transfer function of bioamplifier with active feedback.

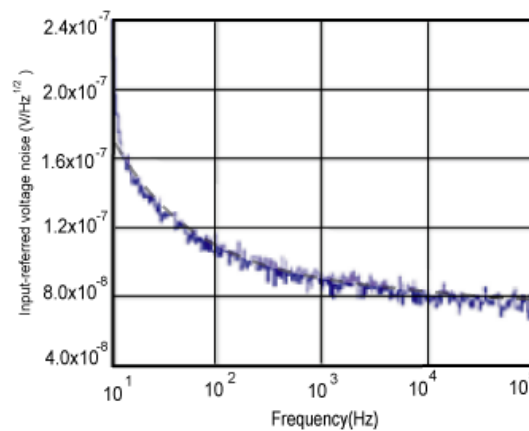


Fig. 9. Measured (solid) and simulated (dashed) input-referred noise voltage densities of bioamplifier.

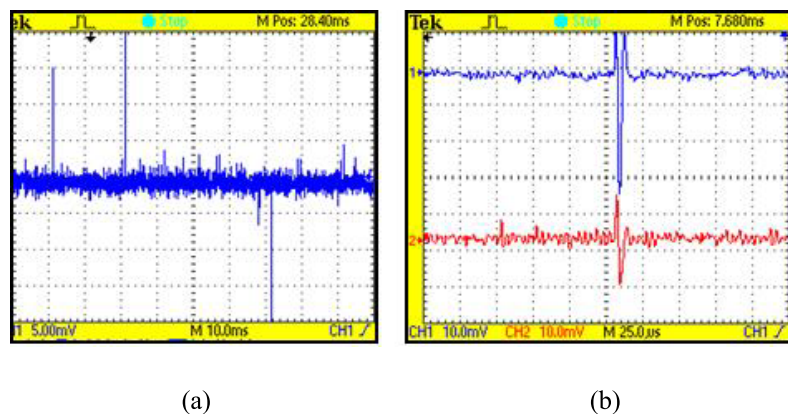


Fig. 10. Measured biopotential from in-vivo extracellular spike of the rat brain. (a) Sample of a biopotential recorded. (b) The output signal of bioamplifier.

The original biopotential signal and output signal of bioamplifier is shown in Fig. 10 by the connected oscilloscope.

Table I. Performance Characteristic of Bioamplifier

	Gain (dB)	BW (Hz)	NEF	Noise (μV_{rms})	Power	Area (mm^2)
[1]	39.5	0.025-7.2k	4	2.2	16 μA @ $\pm 2.5\text{V}$	0.160
[2]	49.5	98.4-9.1k	4.9	5.6	4.67 μA @1.8V	0.050
[3]	40.85	45-5.32k	2.67	3.06	2.7 μA @2.8V	0.160
[4]	39.3	0.001-5k	19.4	7.8	38.3 μA @ $\pm 1.5\text{V}$	0.107
[5]	39.3 45.6	0.015-700	4.9	3.6	8 μA @ $\pm 1.7\text{V}$	0.201
[6]	43.5	0.001-5k	10	3.66	22 μA	0.05
This work	42	13-10.5k	3.1	5.7	2 μA @ $\pm 1.5\text{V}$	0.052

5 Conclusion

We have presented a novel bioamplifier with the active feedback architecture for DC offset suppression function and trade-off design inside low-noise, low-power, and area efficient. This structure features the small feedback capacitors and less gain variation. The experiment results of the bioamplifier with a midband gain of 42 dB have the input-referred noise of 5.7 μV_{rms} over 13 Hz–10.5 kHz. The full differential bioamplifier with the small chip area as single-end output amplifier has high immunity to interference. The neural interface chip of 16-channel readout microsystem is implemented using fully-CMOS process techniques. The complete multichannel recording system including bioAmp, the spike detector, and array correlated double sampling (CDS) module with milliwatt power dissipation. Other connection block require analog multiplexer, comparator, and ADC unit and so on.

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